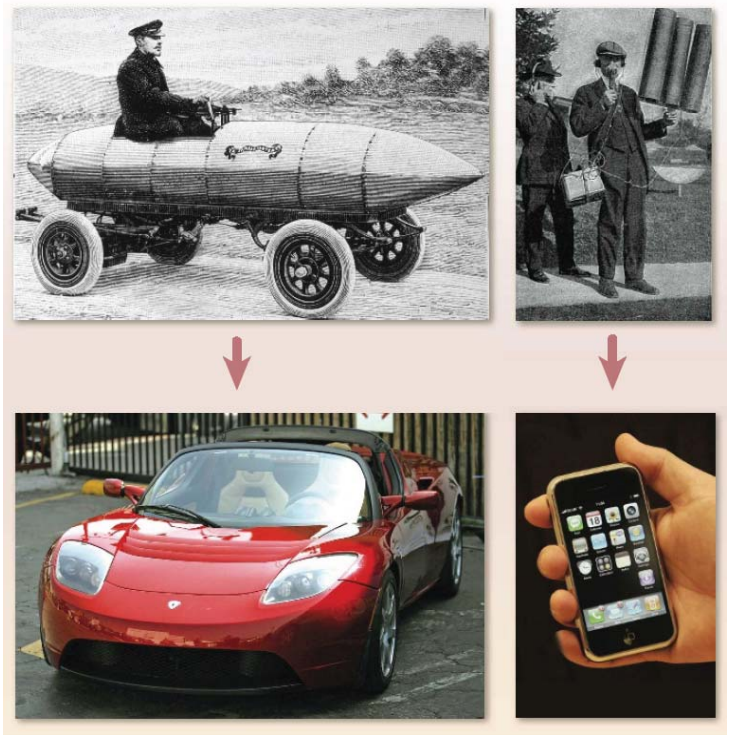
화학특강 (2011년 2학기)
제6장 전기화학과 그 응용: 태양 전지와 연료 전지

중요 학습 내용

- 전지의 화학적 원리
- 전지의 종류
- 태양전지의 원리와 염료감응형 태양전지
- 연료전지의 원리

Building Better Batteries

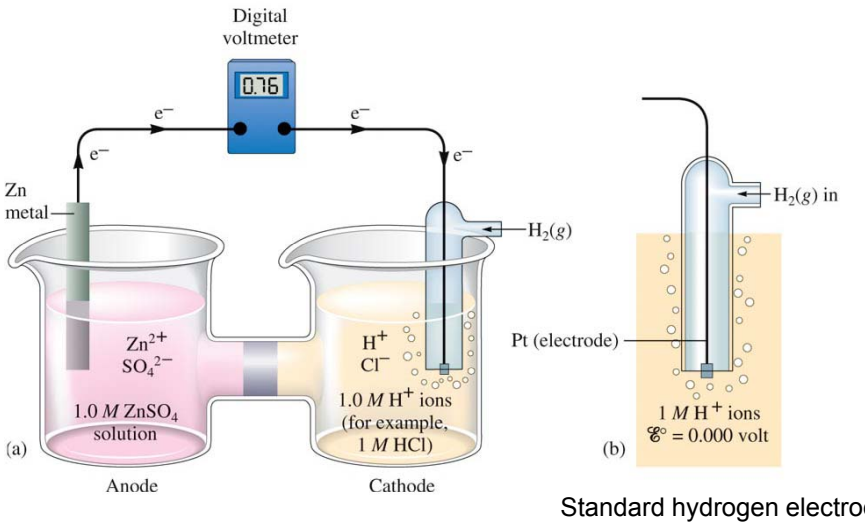


Armand & Tarascon, Nature 451, 652 (2008)

KNU

Standard reduction potential

- Standard reduction potential



Standard hydrogen electrode (SHE)

Anode: $Zn \rightarrow Zn^{2+} + 2e^-$

Cathode : $2H^+ + 2e^- \rightarrow H_2$

Standard Reduction Potentials

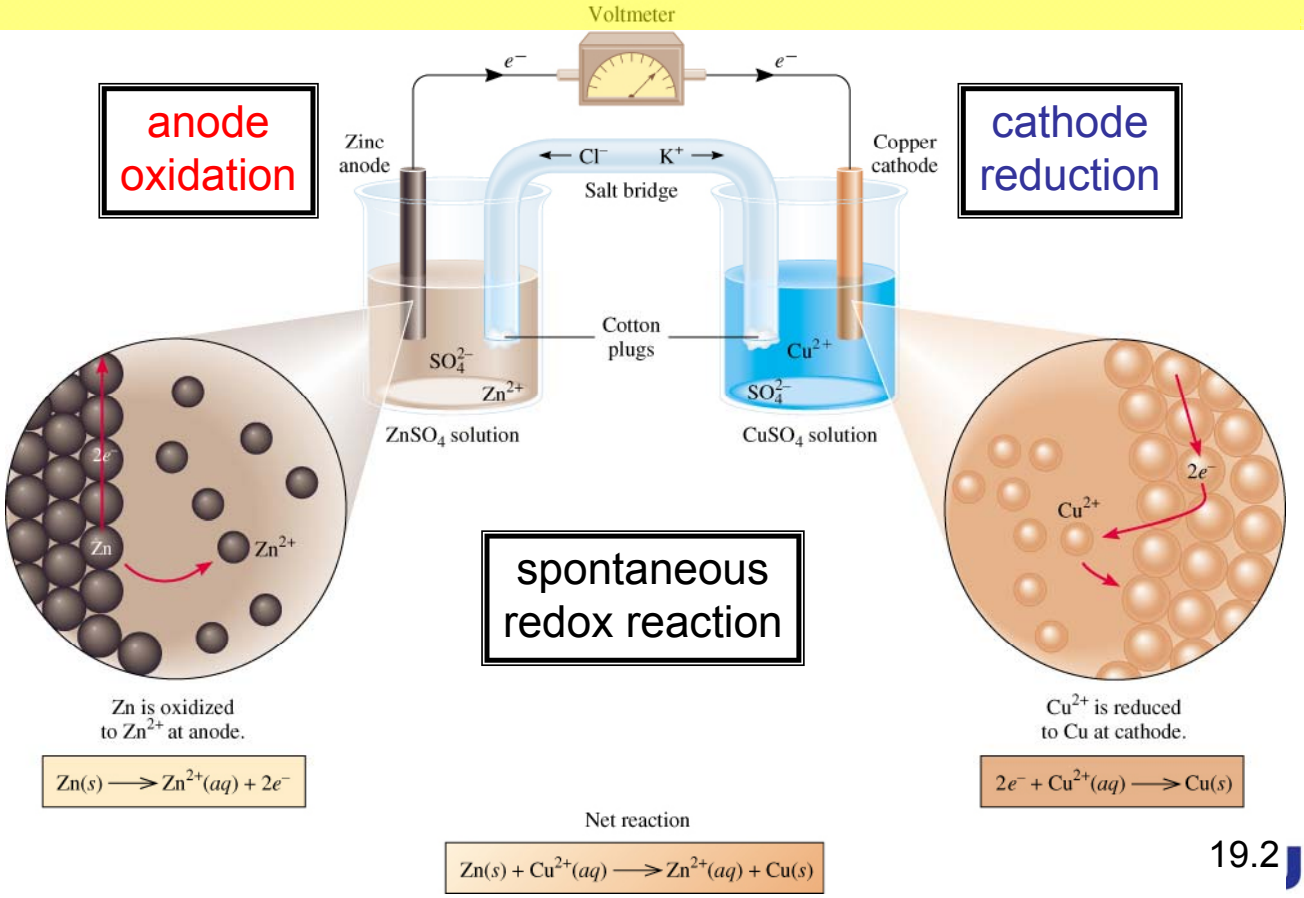
TABLE 17.1 Standard Reduction Potentials at 25°C (298 K) for Many Common Half-Reactions

Half-Reaction	e° (V)	Half-Reaction	e° (V)
$F_2 + 2e^- \rightarrow 2F^-$	2.87	$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$	0.40
$Ag^{2+} + e^- \rightarrow Ag^+$	1.99	$Cu^{2+} + 2e^- \rightarrow Cu$	0.34
$Co^{3+} + e^- \rightarrow Co^{2+}$	1.82	$Hg_2Cl_2 + 2e^- \rightarrow 2Hg + 2Cl^-$	0.27
$H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$	1.78	$AgCl + e^- \rightarrow Ag + Cl^-$	0.22
$Ce^{4+} + e^- \rightarrow Ce^{3+}$	1.70	$SO_4^{2-} + 4H^+ + 2e^- \rightarrow H_2SO_3 + H_2O$	0.20
$PbO_2 + 4H^+ + SO_4^{2-} + 2e^- \rightarrow PbSO_4 + 2H_2O$	1.69	$Cu^{2+} + e^- \rightarrow Cu^+$	0.16
$MnO_4^- + 4H^+ + 3e^- \rightarrow MnO_2 + 2H_2O$	1.68	$2H^+ + 2e^- \rightarrow H_2$	0.00
$2e^- + 2H^+ + IO_4^- \rightarrow IO_3^- + H_2O$	1.60	$Fe^{3+} + 3e^- \rightarrow Fe$	−0.036
$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$	1.51	$Pb^{2+} + 2e^- \rightarrow Pb$	−0.13
$Au^{3+} + 3e^- \rightarrow Au$	1.50	$Sn^{2+} + 2e^- \rightarrow Sn$	−0.14
$PbO_2 + 4H^+ + 2e^- \rightarrow Pb^{2+} + 2H_2O$	1.46	$Ni^{2+} + 2e^- \rightarrow Ni$	−0.23
$Cl_2 + 2e^- \rightarrow 2Cl^-$	1.36	$PbSO_4 + 2e^- \rightarrow Pb + SO_4^{2-}$	−0.35
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$	1.33	$Cd^{2+} + 2e^- \rightarrow Cd$	−0.40
$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$	1.23	$Fe^{2+} + 2e^- \rightarrow Fe$	−0.44
$MnO_2 + 4H^+ + 2e^- \rightarrow Mn^{2+} + 2H_2O$	1.21	$Cr^{3+} + e^- \rightarrow Cr^{2+}$	−0.50
$IO_3^- + 6H^+ + 5e^- \rightarrow \frac{1}{2}I_2 + 3H_2O$	1.20	$Cr^{3+} + 3e^- \rightarrow Cr$	−0.73
$Br_2 + 2e^- \rightarrow 2Br^-$	1.09	$Zn^{2+} + 2e^- \rightarrow Zn$	−0.76
$VO_2^+ + 2H^+ + e^- \rightarrow VO^{2+} + H_2O$	1.00	$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$	−0.83
$AuCl_4^- + 3e^- \rightarrow Au + 4Cl^-$	0.99	$Mn^{2+} + 2e^- \rightarrow Mn$	−1.18
$NO_3^- + 4H^+ + 3e^- \rightarrow NO + 2H_2O$	0.96	$Al^{3+} + 3e^- \rightarrow Al$	−1.66
$ClO_2 + e^- \rightarrow ClO_2^-$	0.954	$H_2 + 2e^- \rightarrow 2H^-$	−2.23
$2Hg^{2+} + 2e^- \rightarrow Hg_2^{2+}$	0.91	$Mg^{2+} + 2e^- \rightarrow Mg$	−2.37
$Ag^+ + e^- \rightarrow Ag$	0.80	$La^{3+} + 3e^- \rightarrow La$	−2.37
$Hg_2^{2+} + 2e^- \rightarrow 2Hg$	0.80	$Na^+ + e^- \rightarrow Na$	−2.71
$Fe^{3+} + e^- \rightarrow Fe^{2+}$	0.77	$Ca^{2+} + 2e^- \rightarrow Ca$	−2.76
$O_2 + 2H^+ + 2e^- \rightarrow H_2O_2$	0.68	$Ba^{2+} + 2e^- \rightarrow Ba$	−2.90
$MnO_4^- + e^- \rightarrow MnO_4^{2-}$	0.56	$K^+ + e^- \rightarrow K$	−2.92
$I_2 + 2e^- \rightarrow 2I^-$	0.54	$Li^+ + e^- \rightarrow Li$	−3.05
$Cu^+ + e^- \rightarrow Cu$	0.52		

Increasing strength as oxidizing agent

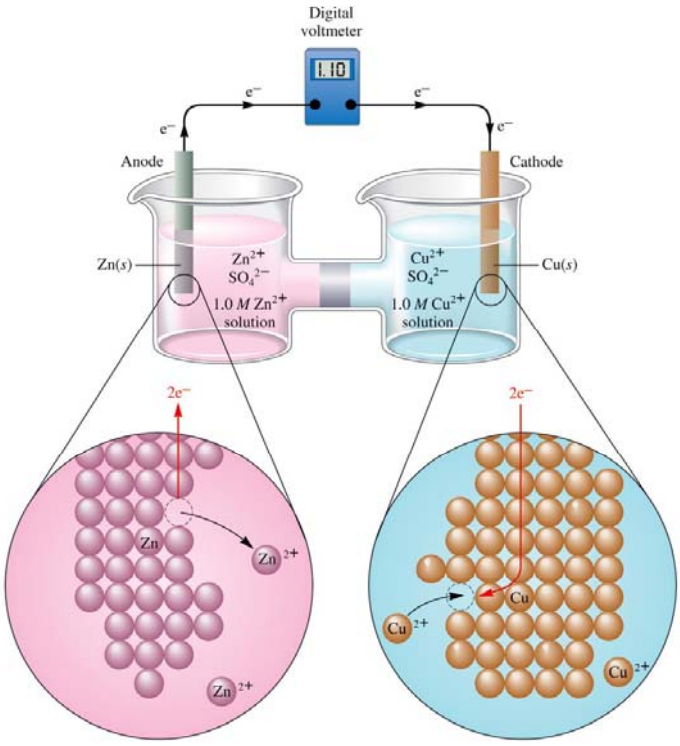
Increasing strength as reducing agent

Galvanic Cells



19.2

Galvanic Cells



Anode: $Zn \rightarrow Zn^{2+} + 2e^-$

Cathode : $Cu^{2+} + 2e^- \rightarrow Cu$

$Zn(s)|Zn^{2+}(aq)||Cu^{2+}(aq)|Cu(s)$

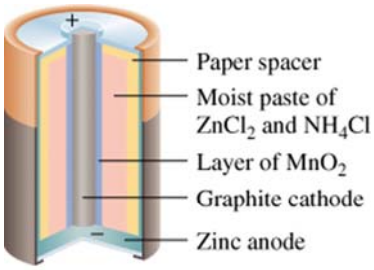
$$E^\circ = E^\circ(\text{cathode}) - E^\circ(\text{anode})$$
$$= 0.34\ \text{V} - (-0.76\ \text{V}) = 0.10\ \text{V}$$

Batteries

- Battery: a galvanic cell, or a series of combined galvanic cells, that can be used as a source of direct electric current at a constant voltage.

  - The dry cell battery
  - The mercury battery
  - The lead storage battery
  - The solid-state lithium battery
  - Fuel Cells

The Dry Cell Battery



Leclanché cell

Anode: $Zn\ (s) \longrightarrow Zn^{2+}\ (aq) + 2e^-$

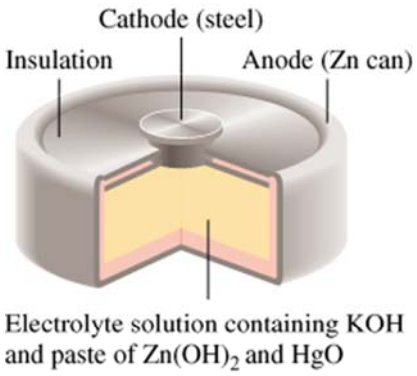
Cathode: $2NH_4^+\ (aq) + 2MnO_2\ (s) + 2e^- \longrightarrow Mn_2O_3\ (s) + 2NH_3\ (aq) + H_2O\ (l)$

Alkaline dry cell: the solid NH_4Cl is replaced with KOH or NaOH.

Anode: $Zn\ (s) + OH^-(aq) \longrightarrow ZnO\ (s) + H_2O\ (l) + 2e^-$

Cathode: $2MnO_2\ (s) + H_2O\ (l) + 2e^- \longrightarrow Mn_2O_3\ (s) + 2OH^-\ (aq)$

The Mercury Battery

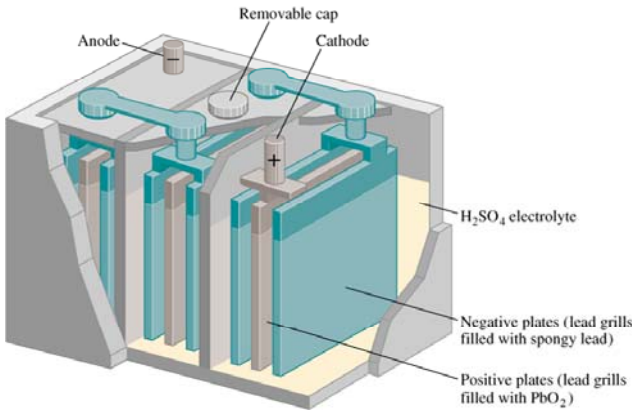


Anode: $Zn(Hg) + 2OH^-(aq) \longrightarrow ZnO(s) + H_2O(l) + 2e^-$

Cathode: $HgO(s) + H_2O(l) + 2e^- \longrightarrow Hg(l) + 2OH^-(aq)$

$$Zn(Hg) + HgO(s) \longrightarrow ZnO(s) + Hg(l)$$

The Lead Storage Battery

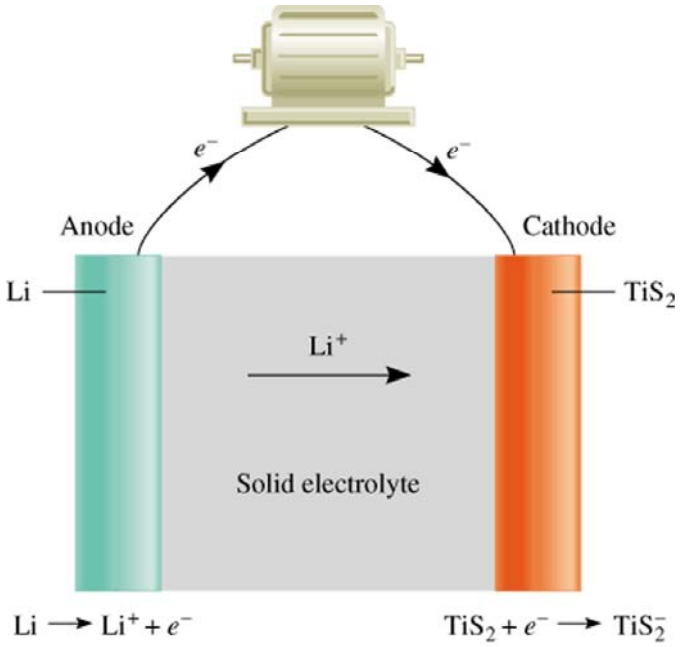


Anode: $Pb(s) + SO_4^{2-}(aq) \longrightarrow PbSO_4(s) + 2e^-$

Cathode: $PbO_2(s) + 4H^+(aq) + SO_4^{2-}(aq) + 2e^- \longrightarrow PbSO_4(s) + 2H_2O(l)$

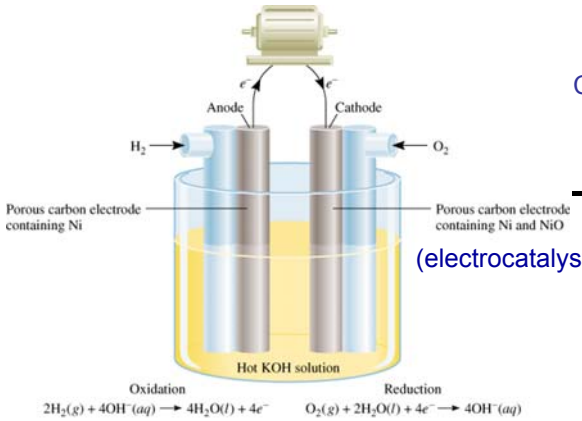
$$Pb(s) + PbO_2(s) + 4H^+(aq) + 2SO_4^{2-}(aq) \longrightarrow 2PbSO_4(s) + 2H_2O(l)$$

The Solid-State Lithium Battery



Fuel Cells

- Fuel cells: a electrochemical (=galvanic) cell that requires a continuous supply of reactants to keep functioning
- Conversion of fossil fuel into electrical energy is a highly inefficient process ~ only 40 %



Anode: $2H_2\,(g) + 4OH^-\,(aq) \longrightarrow 4H_2O\,(l) + 4e^-$

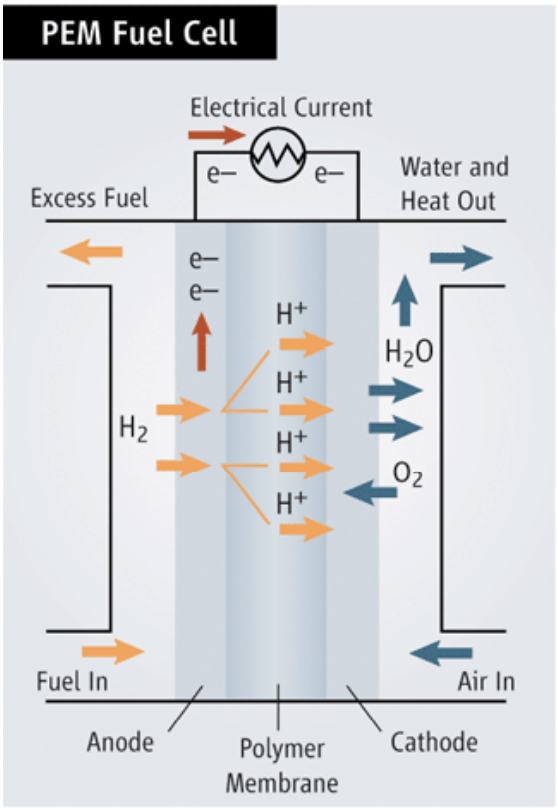
Cathode: $O_2\,(g) + 2H_2O\,(l) + 4e^- \longrightarrow 4OH^-\,(aq)$

$$2H_2\,(g) + O_2\,(g) \longrightarrow 2H_2O\,(l)$$

$$E^\circ(\text{cell}) = E^\circ(\text{cathode}) - E^\circ(\text{anode})$$
$$= 0.40\ \text{V} - (-1.23\ \text{V}) = 1.23\ \text{V}$$

A major problem of Fuel cells lies in the lack of cheap electrocatalysts able to function efficiently for long periods of time with contamination.

Proton Exchange Membrane (PEM) in Fuel Cells



At one electrode, hydrogen molecules are stripped of their electrons, which are then sent through an external circuit to do work. The leftover protons are channeled through the polymer membrane to another electrode, where they meet up with oxygen and the circulating electrons to produce water.

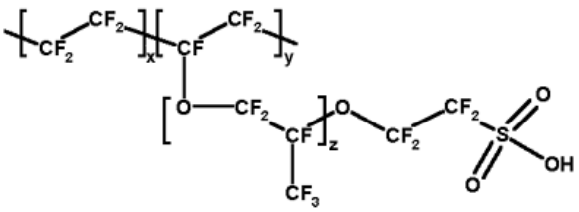
Anode: $2H_2\,(g) \longrightarrow 4H^+\,(aq) + 4e^-$

Cathode: $O_2\,(g) + 4H^+\,(aq) + 4e^- \longrightarrow 2H_2O\,(l)$

$$2H_2\,(g) + O_2\,(g) \longrightarrow 2H_2O\,(l)$$

$$E^\circ(\text{cell}) = E^\circ(\text{cathode}) - E^\circ(\text{anode})$$
$$= 0.0\ \text{V} - (-0.83\ \text{V}) = 1.23\ \text{V}$$

Nafion as a Polymer Membrane



- Nafion
  1. sulfonated tetrafluoroethylene based fluoropolymer copolymer
  2. Made by DuPont
  3. Ionomer → conductivity
  4. Excellent thermal and mechanical stability.
  5. Protons on the SO_3H (sulfonic acid) groups "hop" from one acid site to another.
  6. Pores allow movement of cations but the membranes do not conduct anions or electrons.

Ionomer: a polymer that comprises repeat units of both eleectrically neutral repeating units and a fraction of ionized units (usually no more than 15 percent).

Secondary Electrochemical Cells: Rechargeable Batteries

- Ni-metal hydride batteries (Ni-MH) (M =





- Lithium ion batteries





KNU

Ni-MH batteries

- M = AB_5 where A is a rare earth mixture of lanthanum, cerium, neodymium, praseodymium and B is nickel, cobalt, manganese, and/or aluminium.
- Discharging

Anode: $MH + OH^- \longrightarrow M + H_2O + e^-$

Cathode: $NiO(OH) + H_2O\ (aq) + e^- \longrightarrow Ni(OH)_2 + OH^-$

$$NiO(OH) + MH \longrightarrow Ni(OH)_2 + M$$

- Charging

Anode: $M + H_2O + e^- \longrightarrow MH + OH^-$

Cathode: $Ni(OH)_2 + OH^- \longrightarrow NiO(OH) + H_2O\ (aq) + e^-$

$$Ni(OH)_2 + M \longrightarrow NiO(OH) + MH$$

KNU

Modern Lithium Ion Batteries

- Discharging

Anode: $Li_xC_6 \longrightarrow xLi^+ + 6C + xe^-$

Cathode: $Li_{1-x}CoO_2 + xLi^+ + xe \longrightarrow LiCoO_2$

$$Li_{1-x}CoO_2 + Li_xC_6 \longrightarrow LiCoO_2 + 6C$$

- Charging

Anode: $xLi^+ + 6C + xe^- \longrightarrow Li_xC_6$

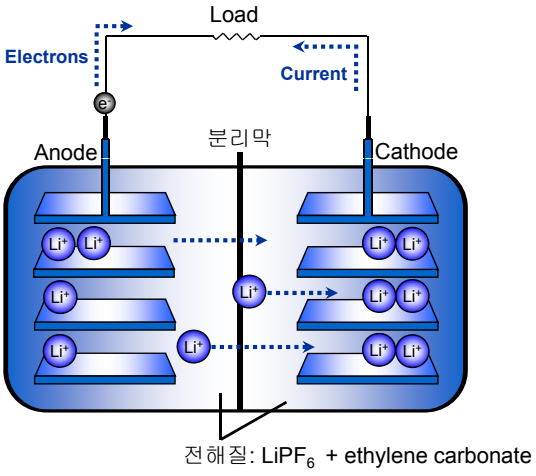
Cathode: $LiCoO2 \longrightarrow Li_{1-x}CoO_2 + xLi^+ + xe^-$

$$LiCoO_2 + 6C \longrightarrow Li_{1-x}CoO_2 + Li_xC_6$$

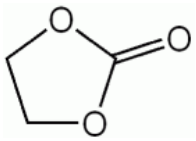
KNU

Mechanism of Lithium Ion Batteries

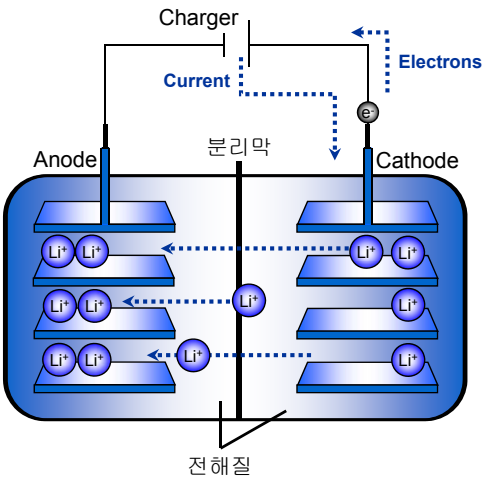
방전 반응 (Discharging)



전해질: $LiPF_6$ + ethylene carbonate



충전 반응 (Charging)



KNU

리튬-이온 2차 전지의 장·단점

- 장 점
  - 에너지 밀도가 높다.
  - 작동 전압이 높다.
  - 무공해
  - 우수한 수명
  - 메모리 효과가 없다.

- 단 점
  - 외부 충격에 약하다.
  - 80℃ 이상의 고열이 가해지면 단락이 발생할 수 있다.
  - 휘발성이 있는 유기성 전해액이 사용된다.
  - 코발트가 지구상에 소량으로 존재

KNU

리튬-이온 2차 전지의 응용

- **휴대폰, MP3, 노트북 등 휴대용 IT기기**







- **HEV (Hybrid Electric Vehicle)**



KNU

전지 개발의 최근 이슈

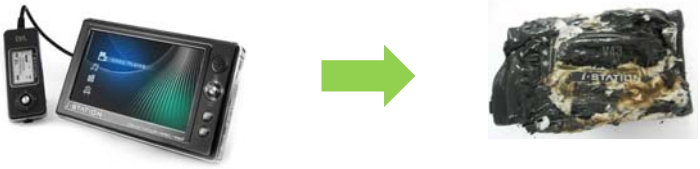
- **리튬폴리머 전지의 개발**
  - 전기 전도도가 우수한 폴리머 전해질 사용
  - 리튬-이온 전지보다 안전
  - 다양한 모양의 전지를 개발



- Swelling 현상 (임신전지)



- 전지 폭발



KNU

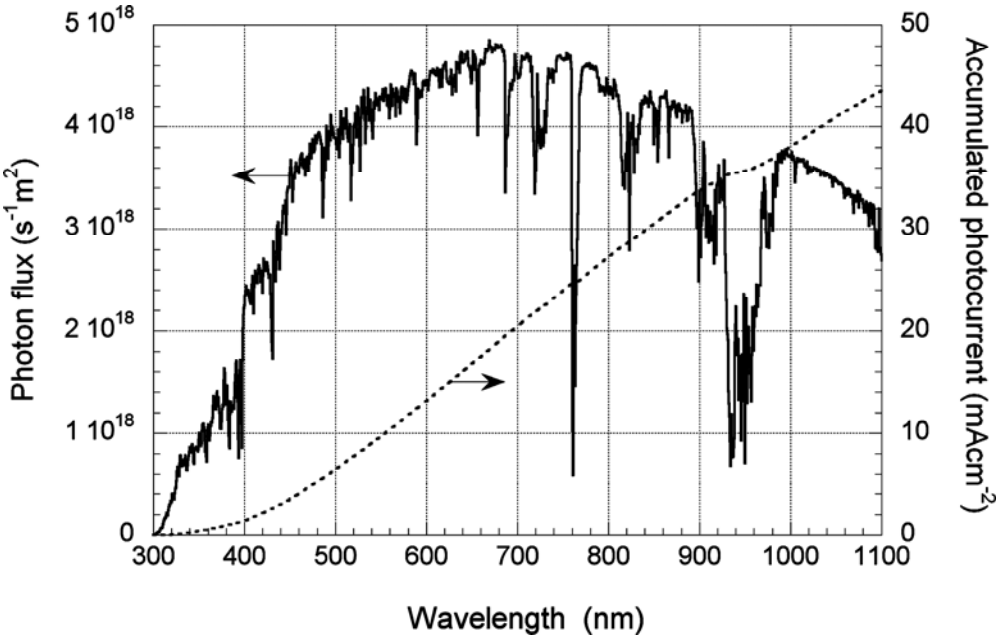
Sun, energy, and Solar Cells

- The mean global energy consumption rate: 13 TW in 2000 → the 28 TW in 2050 in projection
- Solar energy striking earth surface: 1.7×10^5 TW → 600 TW available for an energy source
- The first generation solar cells
  - crystalline silicon (c-Si, mc-Si, etc) based solar cells
  - production cost of $\$3.0/W_p$ compared to that of $\$0.5/W_p$ from $\$0.5/W_p$
  - energy conversion efficiency of 15 % in the lab.
  - energy payback time of 1.5 and 2 years
  - Silicon-based systems make up around 90% of the current photovoltaic market
- The second generation solar cells
  - amorphous silicon (a-si), CdTe, CIGS (copper indium gallium dislenide) thin-film based solar cells
  - energy conversion efficiency of 15 - 20 % in the lab.
  - energy payback time more than 1 year
- The third generation solar cells
  - to deliver electricity at a large scale competitive price, that is, less than \$0.5/Wp.
  - to achieve conversion efficiencies above 31 %

W_p (peak watt): The maximum rated output of a photovoltaic device, such as a solar cell or array, under standardized test conditions, usually 1000 Watt/m^2.

KNU

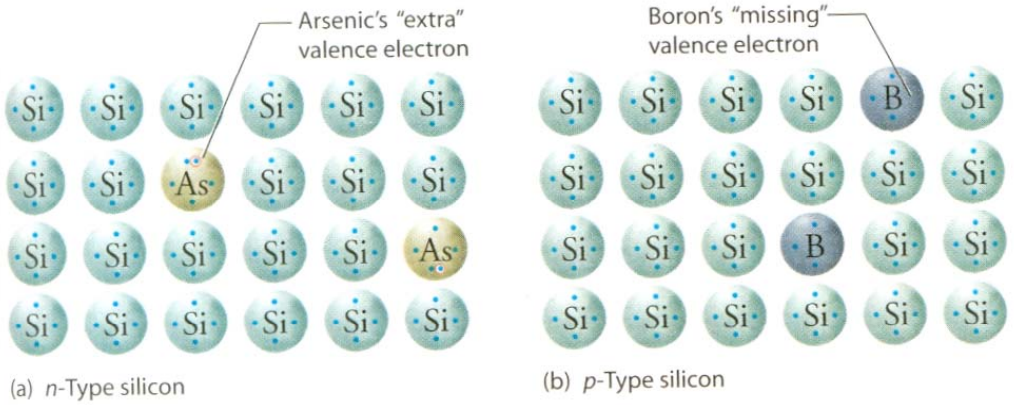
Solar energy



KNU

Solar Cells: A Photovoltaic Cell

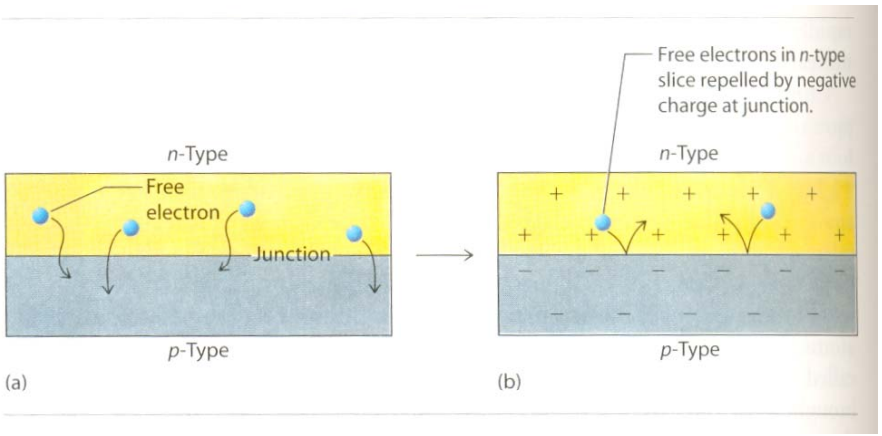
- Photovoltaic cells: a device that converts the energy of sunlight into electricity.
- Semiconductor-based solar cells:



KNU

Silicon-Based Solar Cells

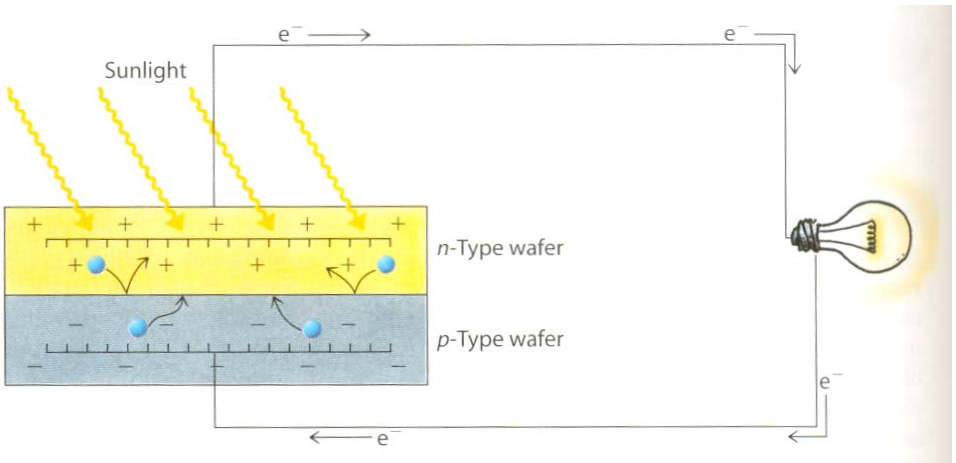
- p-n junction



- (a) Initial, free electrons migrate from the n-type silicon to the p-type silicon.
- (b) Quickly, however, charge buildup at the junction prevents the continued flow of electrons.

Silicon-Based Solar Cells

- Free electrons in the n-type wafer are repelled by the positive charge of the p-type wafer. Free electrons in the p-type wafer, however, are attracted to the positive charge of the n-type wafer. The energy of sunlight gets these electrons moving to the point they are pushed into an external circuit in a unidirectional fashion.



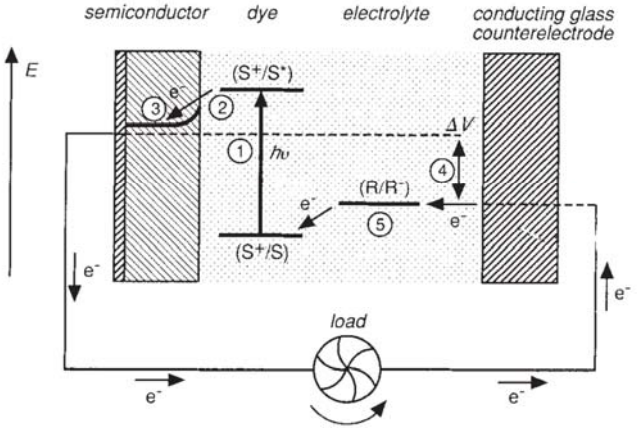
- The energy conversion efficiency of silicon-based solar cells are about 15 %.

Dye-Sensitized Solar Cells (DSSC)

- 1970-1980, a monolayer of dye molecules on a flat surface of ZnO electrode can only absorb up to 1 % of the incident light.
- Nanoporous dye-sensitized TiO_2 electrodes with a high roughness dramatically increased the light harvesting efficiency and in 1991 solar cells of 7 % efficiency were introduced by Grätzel group
- Dye-sensitized solar cells (DSSC) are also called Grätzel cells.



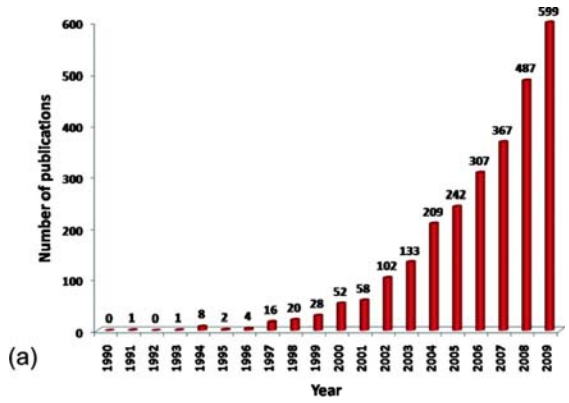
Michael Grätzel
Professor at the Institute of Physical Chemistry,
Swiss Federal Institute of Technology,
Lausanne, Switzerland.

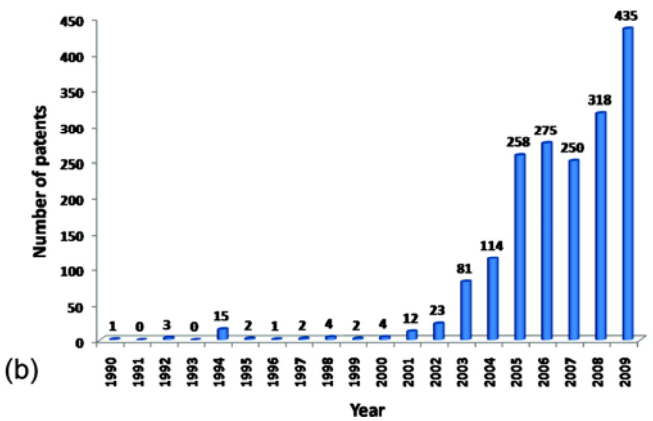


Gratzel et al., Nature **353**, 737 (1991)

KNU

Dye-Sensitized Solar Cells (DSSC)

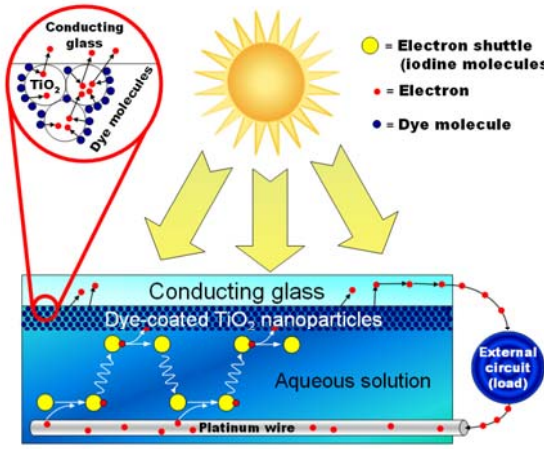


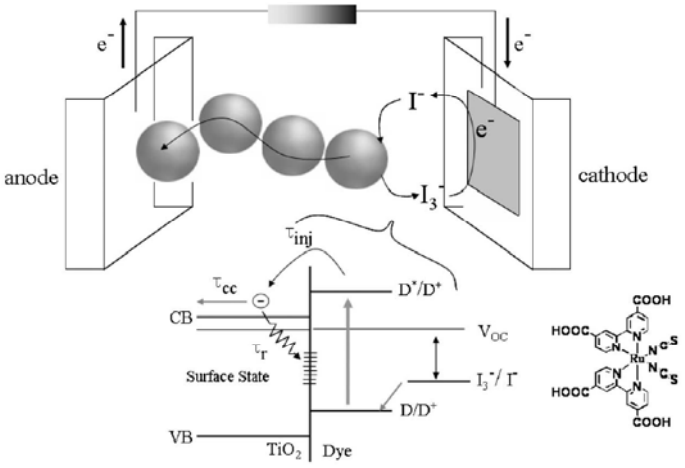


KNU

Dye-Sensitized Solar Cells (DSSC)

- A schematic of the dye-sensitized solar cell

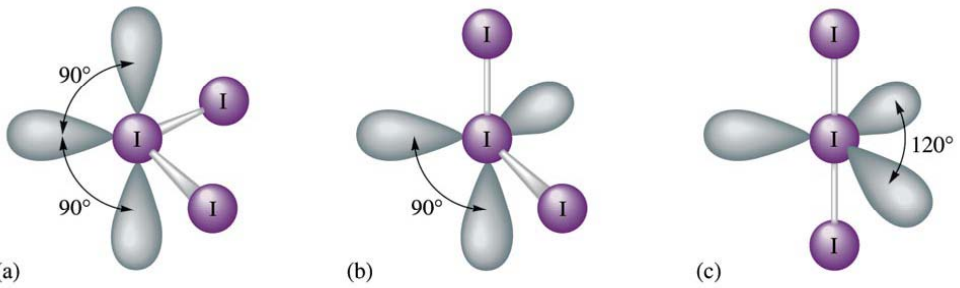




KNU



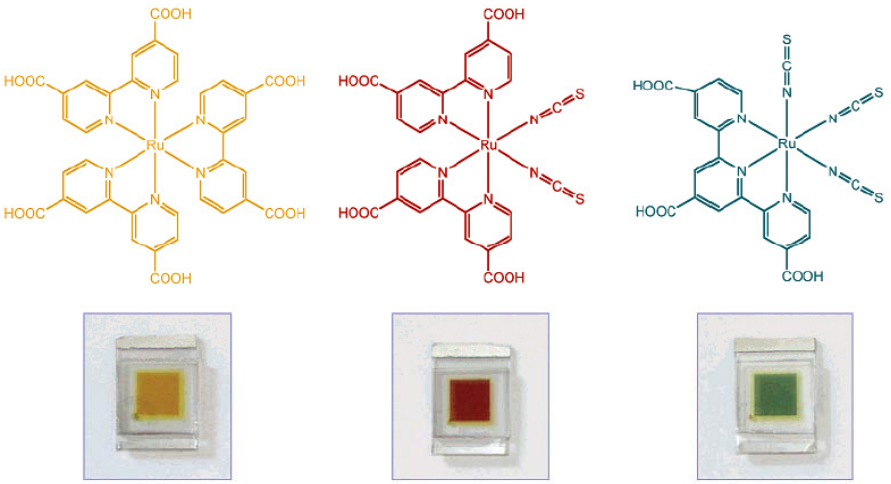
(예제 2) 다음 중 I_3^-의 분자구조는?



(c)의 구조에서 비공유 전자쌍들의 거리가 가장 커서 비공유 전자쌍들간의 반발력이 최소가 되므로 I_3^-는 (c)의 선형 구조를 가진다.

Dye-Sensitized Solar Cells (DSSC)

- Ruthenium-based dyes

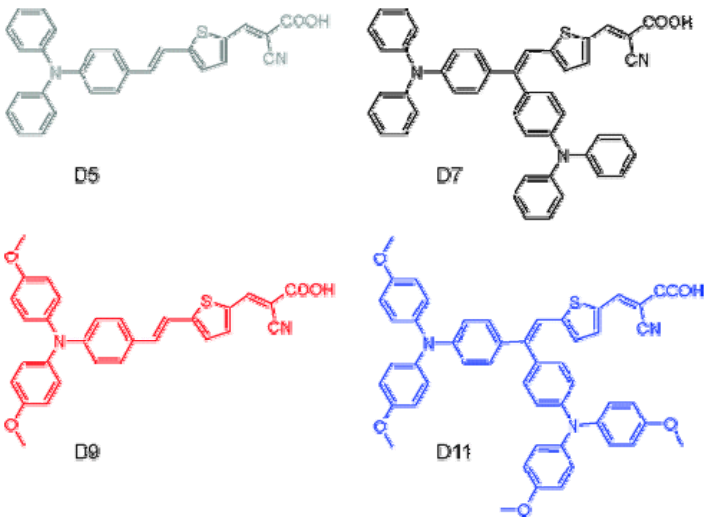


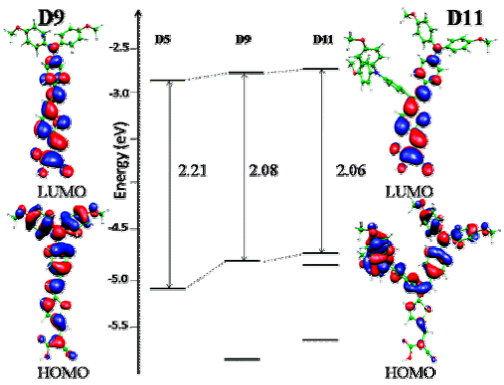
Ruthenium sensitizers RuL_3 (yellow) cis-$RuL_2(NCS)_2$ (red) and $RuL'(NCS)_3$ (green)
where L = 2,2'-bipyridyl-4,4'-dicarboxylic acid and L' = 2,2',2''-terpyridyl-4,4',4''-tricarboxylic acid.

KNU

Dye-Sensitized Solar Cells (DSSC)

- Organic molecule-based dyes





Hgberg et al., JACS **6259**, 130 (2008)

KNU

Strength and weakness of DSSC

- Strength
  - low production cost
  - low energy payback time (< 1 year)
  - enhanced performance under real outdoor conditions - better perfomance under diffuse light conditions and at higher temperatures.
  - possibilities to design solar cells with a large flexibility in shape, color, and transparency.
  - integration into different products opens up new commercial opportunities.
- Weakness
  - low energy conversion efficiency → 6 – 10 % compared with 15 % of Si-based solar cells
  - the formation of dye aggregates on the semiconductor surface
  - recombination of conduction-band electrons with triiodide.
  - dye degradation under heat and UV light.





KNU